



Introduction to Generative AI & The Agentic Revolution



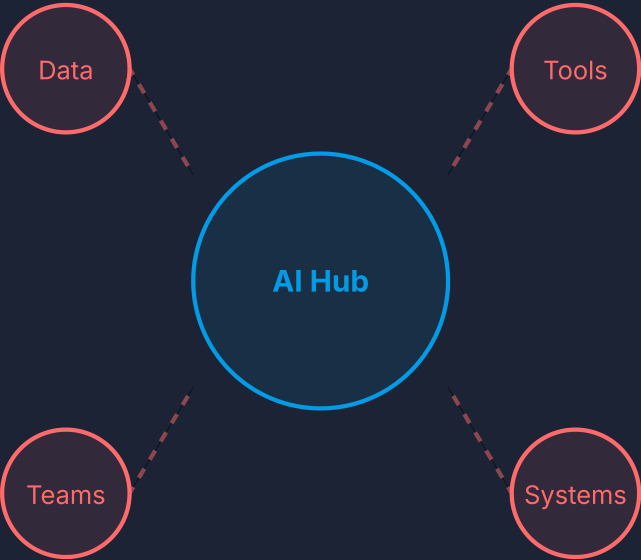
Learning Outcome: By the end of this session, you will be able to explain what Generative AI is, trace its evolution, and articulate exactly where and why it breaks down — setting up the case for agents.

The Problem: Why This Matters Now

Disconnected tools slow decision-making and create costly business gaps

 Critical business impact

The Fragmentation Problem



Disconnected systems create **data silos** and **inefficiencies**

-  **Disconnected Tools** High Impact
Multiple platforms that don't communicate, requiring manual data transfer
-  **Slow Decision-Making** Medium Impact
Data scattered across systems delays critical business decisions
-  **Knowledge Silos** Medium Impact
Critical information locked in individual team repositories
-  **Chatbot Limitations** Low Impact
Current AI assistants sound smart but act powerless
-  **Competitive Pressure** Critical
Competitors ship faster with AI, creating market gaps

73%
Teams report delays

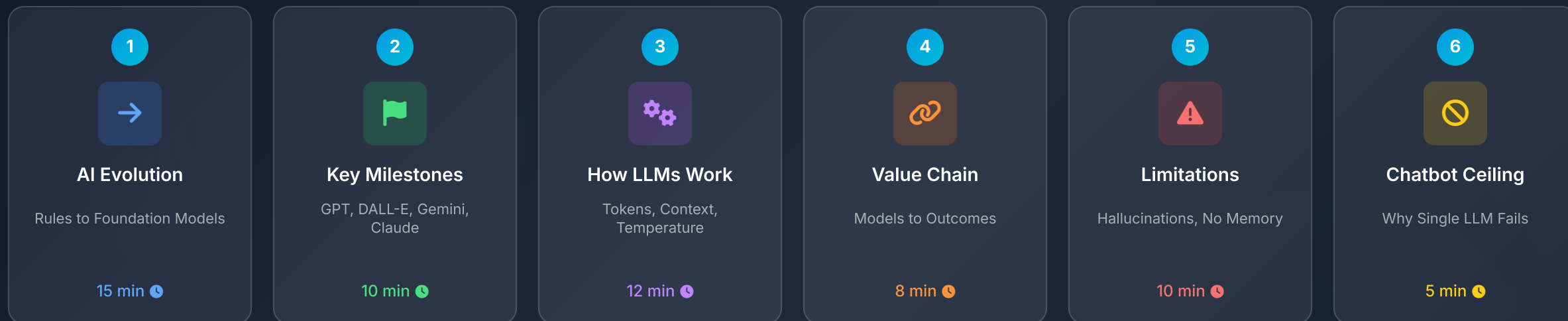
4.2hrs
Daily waste per person

\$2.3M
Annual cost

Every day without AI integration = lost opportunities

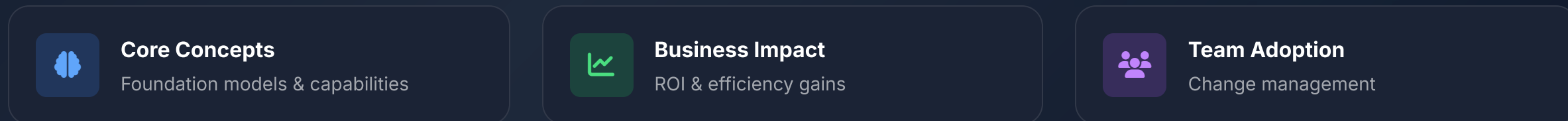
Learning Journey Roadmap

6 sub-topics covering Generative AI fundamentals to Agentic Revolution

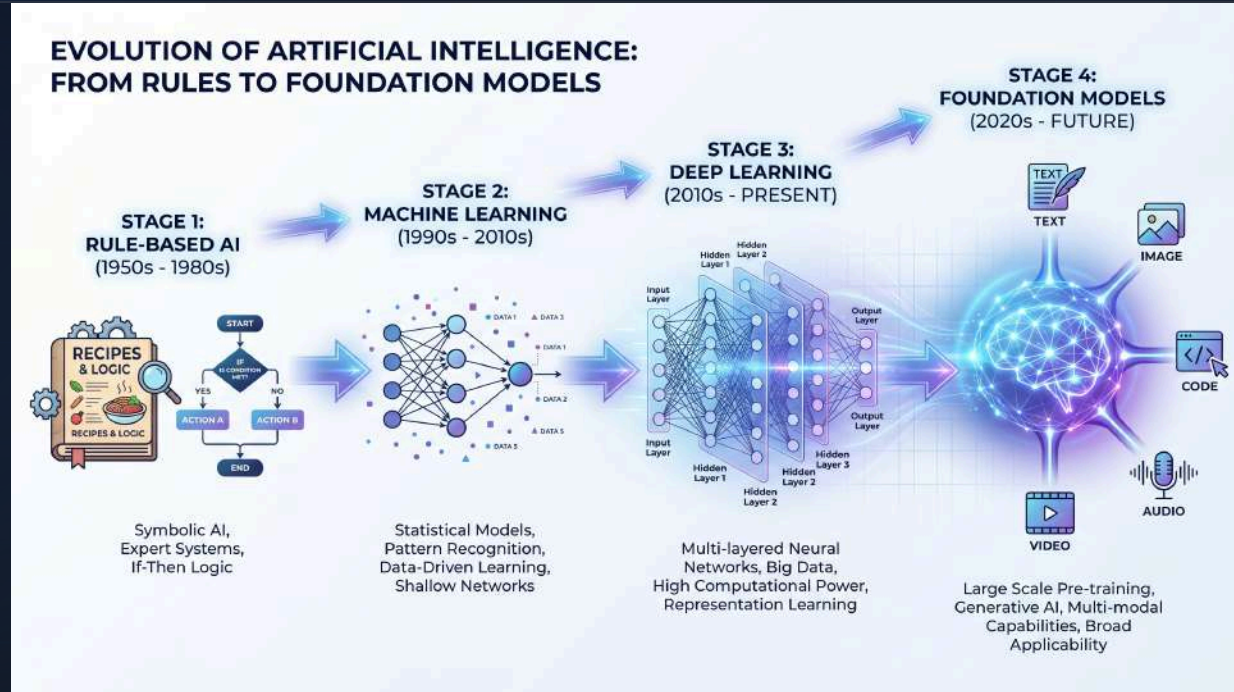


Session Progress

60 minutes total



AI Evolution: Rules → ML → Deep → Foundation



- ✓ **Rule-based:** Fixed if-then recipes.
- ✓ **Machine Learning:** Learns patterns from data.
- ✓ **Deep Learning:** Hidden layers find complex features.

- ✓ **Foundation Models:** Pretrained on internet scale.

💡 *Analogy: Rule-based is a recipe book; GenAI is a master chef.*

In Practice: Handling The Same Query Over Time

 Query Analysis

See how different AI stages process the same customer query: "Where is my refund?"

**Rule-Based**
1980s-2000s

"refund" keyword match

- Exact keyword match
- Fixed response template
- No context understanding

**ML/DL**
2010s

Intent: "refund status"

- Intent recognition
- Pattern matching
- Basic context

**Deep Learning**
2015-2020

Entity: "refund + order"

- Entity extraction
- Context awareness
- Semantic understanding

**Foundation**
2020+

Conversational response

- Full context understanding
- Policy-aware answers
- Natural language

**Thought Prompt**

Think of a task you do at work. Which stage of this evolution handles it today?
→ Consider your daily workflows

**Key Question**

What level of AI sophistication does your current tools provide?

**Impact**

Understanding the evolution helps identify upgrade opportunities

Next: Key Milestones in AI Development

**Response Comparison**

RULE-BASED
"Please check your email for refund status"

ML/DL
"I see you requested a refund for order #12345"

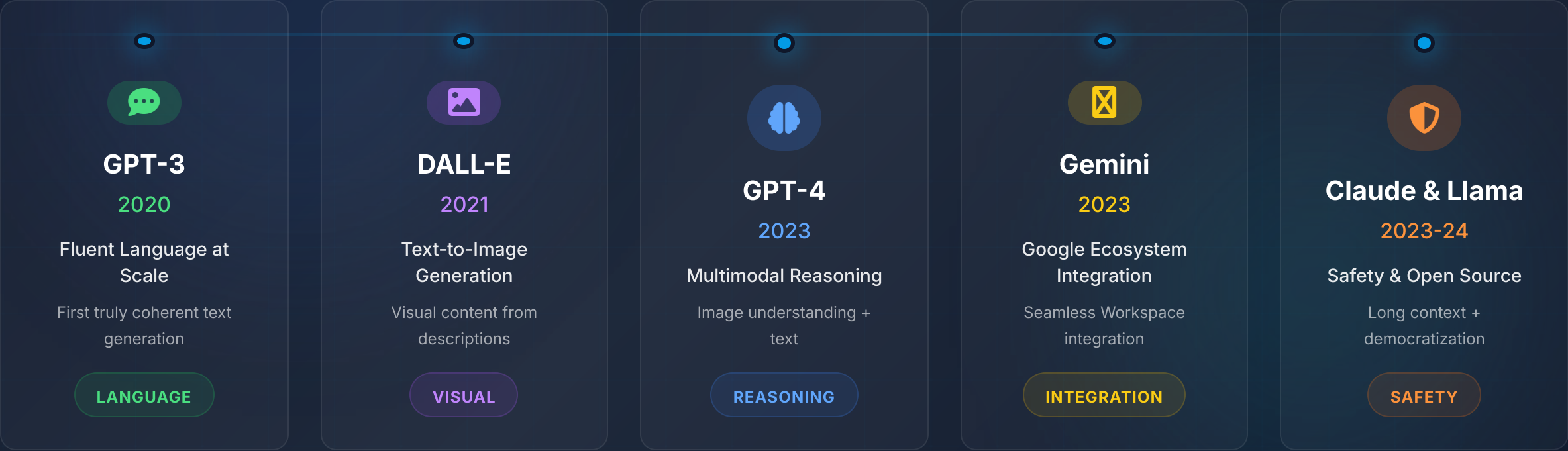
DEEP LEARNING
"Your refund for order #12345 is processing"

FOUNDATION
"Your \$299.99 refund for order #12345 was processed on Jan 15"

Key Milestones: AI Model Evolution

 5 Major Breakthroughs

From GPT-3 to Claude & Llama — what each model unlocked for business



Milestones to Enterprise Use Cases

Mapping AI model capabilities to real business applications



GPT-3 (2020)
Fluent language at scale

Automated drafting



DALL-E (2021)
Text-to-image generation

Marketing visuals



GPT-4 (2023)
Multimodal reasoning

Document analysis



Gemini (2023)
Google ecosystem integration

Workspace copilots



Claude & Llama (2023-2024)
Safety + open-source

Long documents



Automated Drafting
Generate proposals, emails, and content at scale

Marketing



Marketing Visuals
Create campaign imagery and concept mockups

Design



Document Analysis
Extract insights from reports and data

Analytics



Workspace Copilots
Assist with daily tasks and workflows

Productivity



Long Documents
Process large documents with context

Compliance



How LLMs Work: Tokens, Context Windows, Temperature

Understanding the mechanics behind Large Language Models

LLM Processing Flow



Input Prompt
"Explain quantum computing"

Tokens: ~15

↓ TOKENIZATION



Tokenized Input

"Explain" "quantum" "computing"

↓ CONTEXT WINDOW



Context Window
Model's working memory

4,096 tokens

↓ GENERATION



Generated Output
Quantum computing uses quantum bits...



Tokens: The Building Blocks

~4 chars/token

Words broken into sub-word pieces (like LEGO blocks)

Example: "Quantum" → "Qu", "ant", "um"



Context Window: Short-term Memory

4K-32K tokens

The model's working memory / whiteboard

If input exceeds window, oldest tokens are forgotten



Temperature: Creativity Dial

0.0 - 1.0

Controls randomness in output generation



Key Insight: Temperature doesn't change model knowledge, only output style

~4K

MAX TOKENS

0.7

DEFAULT TEMP

~\$0.002

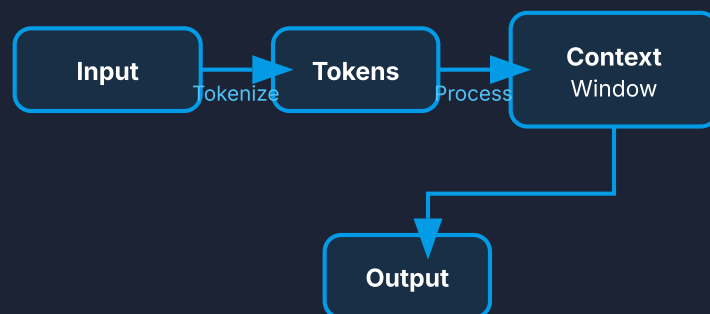
PER 1K TOKENS

Understanding these mechanics helps you optimize prompts

How LLMs Work: Tokens, Context, Temperature

Understanding the mechanics behind Large Language Models

LLM Processing Flow



2,048
CONTEXT WINDOW

4,096
MAX TOKENS

0.7
TEMPERATURE

Tokenization

Words broken into **tokens** like Lego pieces

```
"Hello world" → ["Hello", " world"]  
Tokens: 2 | Characters: 11
```

 Each token ≈ 0.75 words

Context Window

Model's **short-term memory** / whiteboard

Memory Usage 75% used



 4,096 tokens available

Temperature

Creativity dial: **0 (precise)** to **1 (creative)**



Key Insight

Temperature affects **randomness** in output generation

"Lower = more predictable, Higher = more creative"

GenAI Value Chain: Models → Platforms → Applications → Outcomes

Understanding the full stack from foundation models to business value



4-Layer
Architecture



Layer 1: Models

Foundation AI models

- GPT-4
- Claude
- Gemini
- Llama



Layer 2: Platforms

AI infrastructure

- Azure OpenAI
- Vertex AI
- AWS Bedrock
- Hugging Face



Layer 3: Applications

End-user tools

- Copilot
- Chatbots
- Assistants
- Analytics



Layer 4: Outcomes

Business value

- Faster proposals
- Reduced cycle time
- Cost savings
- Productivity gains



Key Insight: Most enterprises operate at the Application layer — they don't need to understand model internals, but they must understand the layers beneath to make good decisions



Integration

Seamless connectivity between layers



Security

Data protection & compliance



Scalability

Enterprise-grade performance

Where EdYoda Sits in the Value Chain

EdYoda Agent Builder: Bridging Models to Applications

 Platform Layer

GenAI Value Chain with EdYoda Positioning

 **Models**
Foundation AI

GPT-4, Claude
Gemini, Llama

 **Platforms**
AI Infrastructure

Azure OpenAI
Vertex AI

 **EdYoda Agent Builder**
App/Platform
Bridge

Memory + Tools
Workflows

 **Outcomes**
Business Value

Productivity
ROI

 **Memory Layer**

Persistent context across
conversations

 **Tool Integration**

Connect to external systems

 **Governance**

RBAC and analytics

EdYoda Advantages

 **No-Code Interface**
Build agents without coding

 **Rapid Deployment**
Deploy across teams quickly

 **Enterprise Security**
SOC 2 compliant

 **Usage Analytics**
Monitor agent performance



Key Insight: EdYoda sits at the Application/Platform layer, bridging foundation models to enterprise use cases with memory, tools, and governance.

Limitations of GenAI: The Critical 2x2 Grid

Understanding where and why AI systems fail in enterprise environments

 4 Critical Issues

Hallucination

 HIGH RISK

Confidence: 95%

Confident wrong answers that appear factual

AI generates plausible-sounding but incorrect information, making it dangerous for critical decisions

 Business Impact: Wrong decisions, reputation damage

Static Knowledge

 MEDIUM RISK

Cutoff: 2023

Knows nothing after training cutoff date

Cannot access real-time information, current events, or recent developments

 Business Impact: Outdated decisions, missed opportunities

No Memory

 MEDIUM RISK

Session only

Forgets everything after conversation ends

Each interaction is independent; no learning from previous conversations

 Business Impact: Inconsistent experiences, repeated work

No Action

 LOW RISK

Text only

Can only produce text, cannot execute tasks

Cannot interact with systems, APIs, or perform actions in the real world

 Business Impact: Manual workarounds, inefficiency

 **Key Insight:** Each limitation has specific business consequences — understanding them is crucial for risk management

78%

\$2.1M

23%

RISK ASSESSMENT REQUIRED BEFORE DEPLOYMENT

Thought Challenge: Where Could This Break?

Identify risks in your processes where GenAI limitations could cause serious problems



Scenario

Regulatory policy just changed

Your company needs to update all customer-facing documentation to reflect new compliance requirements. The AI system was trained before this policy change.



Critical Questions

- Which limitations break compliance?
- How would you mitigate risks?
- What guardrails are essential?

Your Task



Risk Analysis

Identify potential failure points



Hallucination

Confident wrong answers

RISK: HIGH



Static Knowledge

Outdated after training

RISK: MEDIUM



No Memory

Forgets conversations

RISK: MEDIUM



No Action

Only produces text

RISK: LOW

Mitigation Strategies

- Human Review
- Validation Layer
- Fallback Mechanisms

The Hard Ceiling of Chatbots

Why single LLM calls fall short for complex tasks

6 Failure Points

Complex Request Scenario

Manager asks: "Research top 3 competitors, pull last month's sales data, draft strategy memo, schedule review meeting"



Step 1: Research Competitors

Search web for top 3 competitors

✓ Success



Step 2: Pull Sales Data

Access internal CRM system

✗ Fail

Cannot access internal systems



Step 3: Draft Strategy Memo

Create comprehensive document

✗ Fail

No access to document templates



Step 4: Schedule Meeting

Set up calendar event

✗ Fail

Cannot access calendar API

Where Single LLM Fails

Breaking down the limitations

100% System Access

Cannot interact with internal systems or databases

100% Memory

Forgets text between conversations

100% Tools

Can't execute actions in the world

"Every failure you just saw has a fix. That fix is called an Agent."

Agent Built This Session: GenAI Landscape Explorer

A complete AI agent for answering learner questions about AI concepts, tools, and models

AI Agent

GenAI Landscape Explorer

AI Educator Agent

Purpose Answer AI questions

Language Plain business English

Knowledge Base Curated definitions

Input & Output

Input

Free-text question from learner

Output

Jargon-free explanation with analogy

Capabilities

Definitions

Comparisons

Timeline

Analogies

Agent Architecture

Learner Query

"What is GPT-4?"

System Prompt

Role: GenAI Educator

Knowledge Base

Curated data

LLM Processing

Generate response

Output

Plain language

Input Processing

Parse question, identify intent, extract key terms

Knowledge Retrieval

Query knowledge base for relevant information

Recap & What's Next

Summary of key concepts and preview of the next session

6 Topics Covered

AI Evolution

Rule-based to foundation models

Key Milestones

GPT, DALL-E, Gemini, Claude

How LLMs Work

Tokens, context, temperature

Value Chain

Models to applications

Limitations

Hallucinations, no memory

Chatbot Ceiling

Why single calls fail

Next Session: Now that you know where GenAI breaks down — we answer: what do Agents fix, and when should you use one?

Reflection Prompt

Name one process in your organization that GenAI could improve today:

Type your answer here...

And one where its limitations would make it dangerous to deploy without modification: